ELCRO

Hook & Loop Fastening Systems for Seat Assembly

Company Profile

ELCRO CO., LTD.

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01 Company Overview

Company Name	ELCRO CO., LTD.
Date of Establishment	February 2017
CEO	Sang Hee Lee
Address	39-1, Hwalcheonsaneop-ro, Duseo-myeon, Ulju-gun, Ulsan, Korea
Phone	+82 52-264-9104~5
Main Products	Hook & Loop Fastening Systems (Textile Hook, Plastic Hook, Textile Loop)
Key Clients	Hyundai Transys, Hyundai Industrial, FORD, etc.

ELCRO develops and manufactures automotive hook-and-loop fastening systems engineered for polyurethane seat foam and trim-cover assembly. From mushroom-hook forming to lamination, slitting and precision die-cutting, our Ulsan plant covers the full manufacturing process under ISO 9001:2015 quality management.

02 History

Feb 2017	Developed Korea's widest Mushroom Hook production method (1,440 mm)
Apr 2017	Supply contract with Beijing DAS (China)
Aug 2019	Supply contract with Ford & Changsung Motors
Aug 2022	Patent application for the Twin Lock manufacturing method
Jun 2024	Obtained ISO 9001:2015 Quality Management Certification
Oct 2024	Artificial turf installation applying Mushroom Hook; composite waterproofing method developed
Jan 2025	Twin Lock development completed (self-engaging fastener, no male/female distinction)
Jul 2025	Twin Lock supply contracts in China & India

03 Main Products

J SHAPE · Hook & Loop

J-shaped hook profile for seat foam and interior trim applications.

MUSHROOM · Hook & Loop

Mushroom-head hook strip for direct integration into PU seat foam.

T SHAPE · Hook & Loop

T-shaped hook profile for high-retention fastening.

TWIN-LOCK · Self-engaging

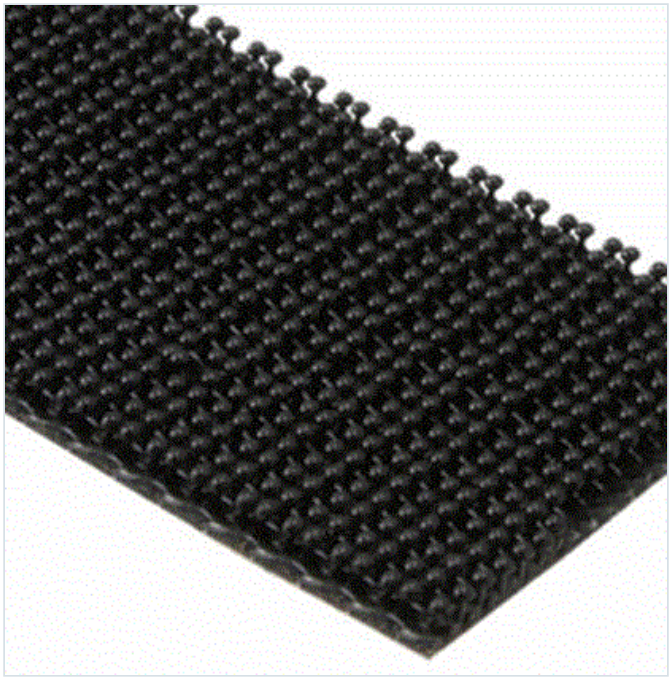
Self-mating fastener for headliners, sunroofs and door trim — no male/female distinction.

TEXTILE LOOP · Loop

Textile loop for general fastening applications.

AUTOMOTIVE LOOP · Loop

Flame-retardant polyester loop sewn into seat trim covers.



Mushroom hook close-up

04 Applications

Automotive Interior Materials

HOOK & LOOP

Trim-cover fastening for seat cushions and seatbacks

TWIN-LOCK

Interior systems including headliners, sunroofs and door trim

05 Assembly Structure

How hook & loop engage in seat assembly.

Component	Role
SEAT COVER	Trim cover with loop sewn in
LOOP	Sewn into cover, mates with hook
HOOK	Mushroom hook integrated into PU foam
SEAT PAD	Polyurethane seat foam

06 Non-Cover Hook Product Composition

Classification	Content
Composition (HOOK)	Polypropylene (Mushroom), Polyester (Base), PET Film, FE Powder, Polyester (Mesh)
Thickness	4.5mm ± 0.1mm
Tolerance	±0.1mm (Length, Width)
Separation Strength	HOOK & LOOP: 15±3 kgf HOOK & SEAT PAD: 22±3 kgf

Shape: Front / Back / Side (standard: 100mm)

07 Cover Hook Product Composition

Classification	Content
Composition (HOOK)	Polypropylene (Mushroom), Polyester (Base)
Composition (Loop)	Nylon Loop
Composition (Cover)	Paper (Base), FE Powder, Nylon (Lylex)
Thickness	4.5mm ± 0.1mm
Tolerance	±0.1mm (Length, Width)
Separation Strength	HOOK & LOOP: 15±3 kgf HOOK & SEAT PAD: 22±3 kgf

Shape: Front / Back / Side (standard: 100mm)

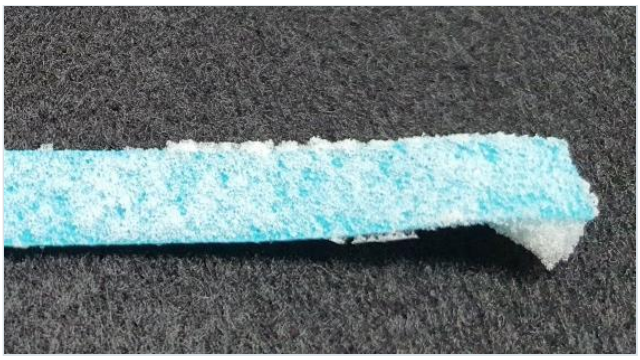
08 Loop Product Composition

Classification	Content
Material	Polyester (phosphorus-based flame retardant yarn)
Product Thickness	3.2mm ± 0.1mm
Product Width	40mm / 38mm / 36mm etc.
Separation Strength	15 kgf ± 2 kgf

Shape: Front / Side / Back (standard: 100mm)

09 Quality Comparison

Item	Competitor Products	ELCRO
Hook & Seat Pad Integration	Approx. 1 mm integrated	**Approx. 35 mm integrated**
Unformed (Void) Prevention	Foaming defects occur	**Rear nylon loop significantly reduces foaming defects**
Hook Rear-Side Integration	Separation when removing cover	**Strong adhesion integration**
Hook Wrinkles	Wrinkles occur (cause seat cover assembly defects)	**No wrinkles**



After foaming — integration (ELCRO)

Hook rear-side integration

10 Separation Strength Test

Unit: kgf | Specimen: 10mm (W) × 100mm (L) | Vertical separation by P/P gauge

Test Item	Result
HOOK & LOOP	**17**
HOOK & PAD	**23**



Tensile tester (Instron 5582)

11 Reliability Test

No	Test Title	Test Standard	Requirement	Result
1	Interior Flammability	MS 300-08	Burn velocity ≤ 80mm/min	Approved
2	Interior Odor	MS 300-34	Grade 3 or higher	Approved
3	Volatile Organic Compounds	MS 300-55	TVOC < 2ppm, HCHO < 2ppm	Approved
4	Regulated Heavy Metals	MS 201-02	Pb, Hg, Cd, Cr6+ not detected	Approved

MS: Hyundai-Kia Motors Engineering Standard (consistent with North American and European motor safety standards)

12 Twin Lock Product Specification

Classification	Content
Composition Layers	A: Polyolefin B: Acrylic Foam C: Acrylic Resin (Adhesive) D: Tape Cover (Release Paper)
Color	BLACK
Thickness	3.6mm ± 0.2mm
Combined Thickness	5.6mm ± 0.3mm
No. of Protrusions	400 (Horizontal 20 × Vertical 20)

Reliability Test

Test Item	Result	Judgment
Flame Retardancy	54mm/min	Pass
Combustibility	Grade 1.5	Pass
Hazardous Substance Prohibition & Reporting	Not detected	Pass
Tensile Strength	20–28 kgf	-

Shape: Front / Back (standard: 25mm × 25mm)

13 Twin Lock Separation Strength Test

Unit: kgf | Specimen: 25mm (W) × 25mm (L) | Instron 5582

1. #400 & #400

Condition	#1	#2	#3	#4
23°C	25.1	26.8	27.5	25.8
-30°C	33.2	32.4	35.5	33.7
120°C	8.3	9.5	8.3	8.7

2. #400 & #250

Condition	#1	#2	#3	#4
23°C	20.8	18.4	19.5	20.2
-30°C	28.7	25.6	23.4	28.2
120°C	7.4	7.6	7.1	7.7

14 Twin Lock & Primer Heat Resistance Test

Specimen: 25mm (W) × 25mm (L) | HMC test specification

Test Method

1. Chamber temperature: 100°C (oven)
2. Weight: 1 kg / 2 kg SPCC steel plate
3. Specimen: RAIL (AL)
4. Apply primer (9960 BTX), attach, test 1 day later
5. 8-hour test

Configuration	Result
#400 & #400	**APPROVED**
#400 & #250	**APPROVED**

15 Twin Lock & Primer 3-Cycle Test

Specimen: 25mm (W) × 25mm (L) | HMC test specification

Test Method (3-Cycle / 1-Cycle standard)

High temperature (120°C) 8 hours → Low temperature (-40°C) 8 hours → Room temperature (23°C/50% RH) 8 hours

Primer	Result
520K	**PASS**
4298UV	**PASS**
9960BTX	**PASS**

Application areas: Side molding, Rail

16 Twin Lock Quality Certification

No	Test Title	Test Standard	Requirement	Result
1	Interior Combustibility	MS 300-08	Combustion speed ≤ 80mm/min	Pass
2	Interior Odor	MS 300-34	Grade 3 or higher	Pass
3	Hazardous Substance Prohibition & Reporting	MS 201-02	-	Pass

MS: Hyundai & Kia Motors Technical Standards (compliant with North American and European automotive safety standards)

17 Twin Lock Applications

Sunroof & Headliner

Self-engaging fastening for interior trim fixation

Quarter Glass

QTR GLASS — glass molding fixation

Door Trim

DOOR TRIM — door interior fastening



Sunroof frame application example

18 Equipment Status

Equipment	Purpose
Thermoforming Machine	Hook forming
Slitting Machine	Slitting to specifications
Lamination / Die-Cutting Machine	Lamination and die-cutting
Extrusion Coating Machine	Film / fabric extrusion coating
Mushroom Forming Equipment	MUSHROOM hook forming
Fabric Cutting Equipment	Fabric cutting to specifications
Finished Product Cutting Equipment	Finished product cutting
Non-Cover Production Equipment #1 / #2	Non-cover hook production
Double / Six-Fold Forming Machine	Multiple forming
Twin-Lock Molding Equipment	Twin-Lock molding
Twin-Lock Post-Processing Equipment	Twin-Lock post-processing

Thank You

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